

- 4 A 28-year study of Magellanic penguins, *Spheniscus magellanicus*, found in Argentina, provides evidence of natural selection.

Magellanic penguins lay their eggs in nests. They use their bills (beaks) to catch prey and feed their chicks (offspring) in the nest. Each breeding pair of penguins uses the same nest each year.

A Magellanic penguin is shown in Fig. 4.1.



Fig. 4.1

- Data were collected for bill size every year from 1983 to 2010.
- Bill size was calculated using the length and depth of the bill.
- Bill size showed variation between the individuals.
- In 1983 all the penguins in one area were tagged.
- All tagged penguins were measured each year and their new chicks were tagged and measured.
- For each year of the study, an estimate of food availability was made.
- A statistical analysis was conducted to quantify whether selection had taken place.

(a) Explain why bill size is an example of continuous variation.

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(b) Statistical analysis of the data showed that selection was **not** significant in most years of the study. However, a significant increase in bill size occurred in four years of the study.

(i) Name the type of selection that occurred in these four years.

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(ii) In these same four years, food availability was low.

Explain how the data for bill size and food availability supports the idea of the 'struggle for existence' seen in natural selection.

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(c) Further investigation showed that, in some years, larger bill sizes of adult males correlated with higher reproductive success.

Reproductive success was measured by the number of chicks that survived per adult each year.

Suggest why larger bill size of adult males correlated with higher reproductive success.

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..... [1]

[Total: 6]